

Notes: Acharya and Mora (JF, 2015)

A Crisis of Banks as Liquidity Providers

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1 Big picture

- **Question.** Can banks continue to provide liquidity during a systemic financial crisis when the banking system itself is perceived as risky?
- **Traditional view.** Banks combine deposit-taking with loan commitments because deposit inflows in market-wide stress can fund commitment drawdowns. This is the Kashyap-Rajan-Stein / Gatev-Strahan liquidity-synergy view.
- **Paper's challenge.** The 2007–2009 crisis was different because banks were themselves at the center of the shock. Depositors did not initially flee toward banks; many fled toward explicitly government-backed instruments such as Treasuries, government money-market funds, FHLB discount notes, and later insured accounts.
- **Main aggregate fact.** From the ABCP freeze in August 2007 through mid-2008, aggregate bank lending growth outpaced deposit growth, producing a loan-to-deposit shortfall. Core deposit inflows only surged after Lehman and after stronger public guarantees.
- **Main bank-level fact.** Banks with larger unused commitments, and hence greater liquidity demand risk, raised deposit rates more during the first phase of the crisis, but experienced weaker deposit growth, widened lending-deposit shortfalls, used liquid buffers, borrowed from government-related sources, and reduced new credit.
- **Core interpretation.** Banks honored outstanding credit lines, but this liquidity provision was not privately self-sustaining. It relied heavily on explicit government and government-sponsored support.
- **Economic takeaway.** Banks' liquidity-production advantage can fail exactly when bank balance sheets are questioned. Deposit-taking and commitment lending are complements in normal stress but can become jointly fragile in a banking-centered crisis.

2 Incremental contribution of the paper

- **Revises the bank liquidity-provider view.** Prior work emphasized that banks are natural liquidity hedges because deposit inflows offset credit-line drawdowns during market stress. Acharya and Mora show that this hedge can break down when the crisis is centered on banks rather than outside the banking sector.
- **Distinguishes aggregate and individual-bank liquidity.** The paper shows both an aggregate loan-to-deposit shortfall for the banking system and bank-level stress among institutions with high unused commitments.
- **Separates liquidity demand risk from solvency risk.** The authors test whether commitments-exposed banks are simply weak banks. They show that liquidity exposure matters separately from non-performing loans, real-estate exposure, and capital ratios.
- **Introduces commitment exposure as a crisis vulnerability.** Unused commitments are not just off-balance-sheet promises; during a funding crisis, they become contingent liquidity demand that can strain balance sheets.
- **Shows the limits of private deposit competition.** Banks with high liquidity demand risk raise deposit rates, but the higher rates are insufficient to generate enough stable deposits.
- **Documents government as liquidity backstop.** The paper emphasizes FHLB advances, Federal Reserve facilities, and deposit-insurance expansions as critical supports that allowed banks to continue honoring promises.
- **Reconciles the 2007–2009 crisis with previous episodes.** By revisiting earlier crises using stress interactions, the paper shows that unused commitments previously predicted deposit inflows, whereas in 2007–2009 the interaction reverses.

- **Policy contribution.** The findings speak directly to Basel III-style liquidity regulation, the pricing of lender-of-last-resort access, deposit-insurance design, and systemic-risk-sensitive liquidity requirements.

3 Data and measurement

3.1 Aggregate funding and stress data

- Weekly bank balance-sheet aggregates come from the Federal Reserve H8 release.
- Deposit flows are aggregated from weekly to monthly or quarterly frequency.
- Market stress is proxied by the VIX and commercial-paper spreads.
- Money-market mutual fund assets and yields are from iMoneyNet.
- The aggregate analysis distinguishes the first phase of the crisis, 2007Q3–2008Q2, from the post-Lehman / government-backstop phase, 2008Q3–2009Q2.

Interpretation: The aggregate data show that the classic flight-to-bank-deposits mechanism was weak at the start of the crisis and only became powerful after public guarantees became more explicit.

3.2 Bank-level panel

- The bank-level analysis uses quarterly Call Report data from 1994 to 2009.
- The unit is a bank organization.
- All regressions include bank fixed effects and quarterly time dummies.
- Standard errors are clustered at the bank organization level.
- The main liquidity-risk measure is lagged unused commitment ratio.

Interpretation: Unused commitments measure potential future funding demand from borrowers. A bank with more unused commitments is more exposed to drawdowns during a credit-market freeze.

3.3 Deposit-rate data

- Deposit rates come from Bank Rate Monitor, a proprietary weekly survey.
- The paper uses rates on checking accounts, core deposits, large time deposits, and CD maturities.
- Figure 1 compares failed and nonfailed banks, showing that weak banks paid higher deposit rates before failure.

Interpretation: Deposit-rate data are important because quantities alone cannot tell whether banks actively tried to raise funds. Higher offered rates reveal bank demand for deposits.

3.4 Key variables

- **Unused Commitment Ratio:** unused commitments divided by assets.
- **Deposit growth:** changes in deposits scaled by lagged assets or by deposit base.
- **Core deposits:** stable deposits excluding large time deposits.

- **Loan-to-deposit shortfall:** lending growth minus deposit growth.
- **Other borrowed money:** nondeposit borrowing, including government and government-sponsored sources such as FHLB advances.
- **Solvency proxies:** nonperforming loans, real-estate loan share, and capital ratio.

Interpretation: The variables distinguish liquidity pressure from solvency pressure and show how liquidity-short banks adjusted margins.

4 Models and empirical specifications

4.1 Bank-level crisis interaction model

The baseline specification has the form

$$Y_{it} = \alpha_i + \delta_t + \beta_0 \text{Commitments}_{i,t-1} + \beta_1 \text{Commitments}_{i,t-1} \times \text{Crisis1}_t + \beta_2 \text{Commitments}_{i,t-1} \times \text{Crisis2}_t + X'_{it} \Gamma + \varepsilon_{it}.$$

- Y_{it} is a deposit rate, deposit growth, loan growth, loan-to-deposit shortfall, liquid asset growth, or borrowing growth.
- Crisis1 is 2007Q3–2008Q2.
- Crisis2 is 2008Q3–2009Q2.
- Bank fixed effects absorb time-invariant bank business models.
- Time dummies absorb aggregate shocks common to all banks.

Interpretation: The coefficients on the interactions show whether commitment-exposed banks behave differently during the crisis relative to normal times.

4.2 Deposit-rate and deposit-growth interpretation

$$\beta_1^{\text{rate}} > 0 \quad \text{and} \quad \beta_1^{\text{deposit}} < 0$$

means banks with more unused commitments raise rates but still experience weaker deposit growth in Crisis1.

Interpretation: This combination is central. It shows that banks were demanding deposits more intensely but could not attract enough private funding.

4.3 Lending and shortfall interpretation

$$\Delta \text{Loans} - \Delta \text{Deposits} > 0$$

for high-commitment banks indicates that credit-line drawdowns and loan growth exceeded deposit funding.

Interpretation: This is the balance-sheet signature of a breakdown in the deposit-commitment liquidity synergy.

4.4 Liquidity adjustment margins

The paper studies whether banks respond by changing liquid assets and borrowing:

$$\Delta \text{LiquidAssets}_{it}, \quad \Delta \text{InterbankBorrowing}_{it}, \quad \Delta \text{OtherBorrowedMoney}_{it}.$$

Interpretation: If private deposits are insufficient, liquidity-short banks must run down liquid buffers, borrow in interbank/repo markets, or use government-related borrowing.

4.5 Previous-crises triple-interaction model

$$Y_{it} = \alpha_i + \delta_t + \beta_1 \text{Commitments}_{i,t-1} \times \text{Stress}_t + \beta_2 \text{Commitments}_{i,t-1} \times \text{Stress}_t \times \text{Crisis}_t + X'_{it}\Gamma + \varepsilon_{it}.$$

Interpretation: The triple interaction tests whether the relationship between commitments, market stress, and deposits/lending differs in the 2007–2009 crisis relative to previous market-stress episodes.

5 Identification and empirical design

5.1 Core identification challenge

High-commitment banks may differ from low-commitment banks in business models, asset portfolios, size, capital, geography, and solvency. The design therefore uses bank fixed effects, time effects, controls, geographic concentration controls, and solvency splits to isolate liquidity demand risk.

5.2 Why unused commitments identify liquidity demand risk

- Unused commitments are promises made before the crisis.
- Drawdowns rise when firms lose access to market funding.
- Banks must fund these drawdowns even if deposits do not arrive.
- Therefore, unused commitments measure ex ante exposure to crisis liquidity demand.

Interpretation: The measure is powerful because it is determined before crisis liquidity needs are realized.

5.3 Separating liquidity from solvency

- Table VIII splits banks by nonperforming loans, real-estate share, and capital ratio.
- Real-estate and weak-capital banks are solvency-risk categories.
- Commitment exposure remains informative within these groups.

Interpretation: The liquidity story is not reducible to the statement that bad banks paid higher rates. Liquidity-demand exposure separately predicts stress.

5.4 Why previous-crisis comparisons matter

In earlier market-stress episodes, unused commitments were associated with deposit inflows and lending growth. In 2007–2009, when bank safety was questioned, the sign changes. This comparison helps identify the mechanism: the problem is not commitments per se, but commitments during a bank-centered crisis.

5.5 Main limitations

- The analysis is observational and cannot randomly assign commitment exposure.
- The bank fixed-effects design controls for time-invariant differences, but time-varying unobservables may remain.
- Some government support channels are difficult to measure perfectly.
- The results are U.S.-focused, though the authors discuss related international evidence.

Interpretation: The evidence is strongest as a consistent set of aggregate and bank-level patterns rather than a single quasi-random experiment.

6 Section-by-section study guide

- **Introduction.** Establishes the puzzle: banks honored credit lines during the crisis, but did they do so using private liquidity or public support?
- **Section I.** Reviews the traditional view of banks as liquidity providers and shows aggregate evidence that the deposit inflow mechanism broke down before Lehman.
- **Section II.A-B.** Develops hypotheses and shows that high-commitment banks raised deposit rates but had weaker deposit growth.
- **Section II.C-D.** Shows that high-commitment banks reduced new credit, ran down liquid assets, and borrowed from government-related sources.
- **Section II.E-F.** Distinguishes liquidity demand risk from solvency risk and effective drawdowns from potential exposure.
- **Section III.** Revisits previous crises and shows that 2007–2009 was different.
- **Conclusion.** Draws policy implications for liquidity regulation, deposit insurance, and lender-of-last-resort design.

7 Tables and figures

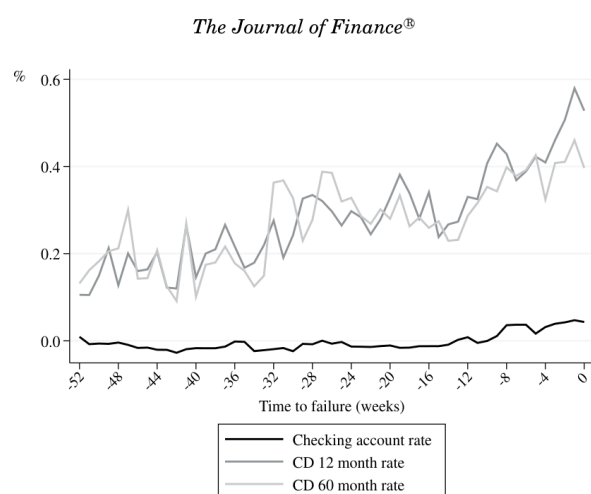
7.1 Figures 1–2: Deposit-rate pressure and money-market fund flows

Read:

- Figure 1 shows that failed and nearly failed banks paid substantially higher CD rates before failure.
- Figure 2 shows that investors moved toward government money-market funds before Lehman and that prime fund yields spiked relative to government funds.
- Together, the figures show both bank-level deposit-price stress and system-wide preference for explicitly government-backed liquidity.

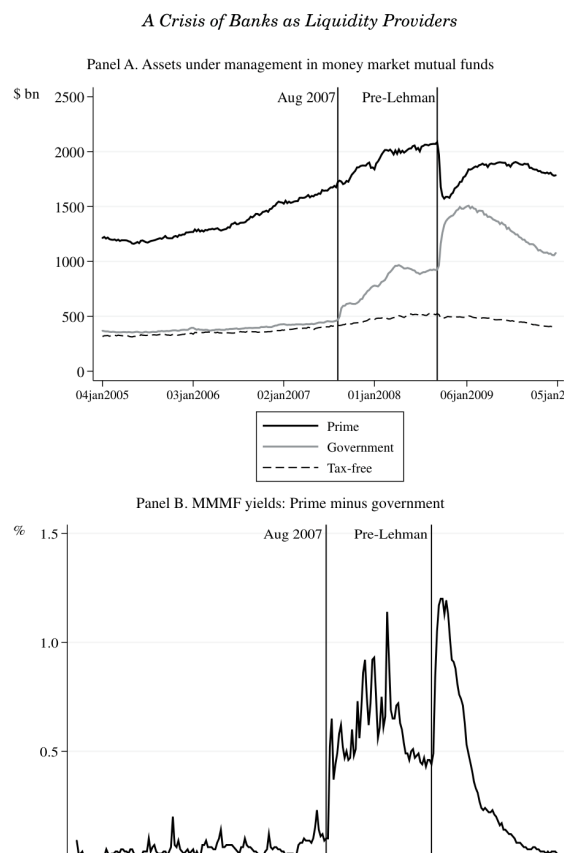
Detailed interpretation: Figure 1 demonstrates that deposit markets disciplined weak banks by requiring higher rates. Figure 2 explains why this mattered in 2007–2008: investors seeking safe liquidity were not automatically moving into bank deposits. They were moving into government-backed money funds and similar instruments.

Economic message: When banks are perceived as risky, their traditional deposit-funding advantage weakens. Higher deposit rates become a symptom of funding stress rather than a sign of abundant deposit supply.



1. **Deposit rates: Failed minus nonfailed banks.** This figure plots the depo

(a) Figure 1: Deposit rates, failed minus nonfailed banks



(b) Figure 2: Money market mutual funds

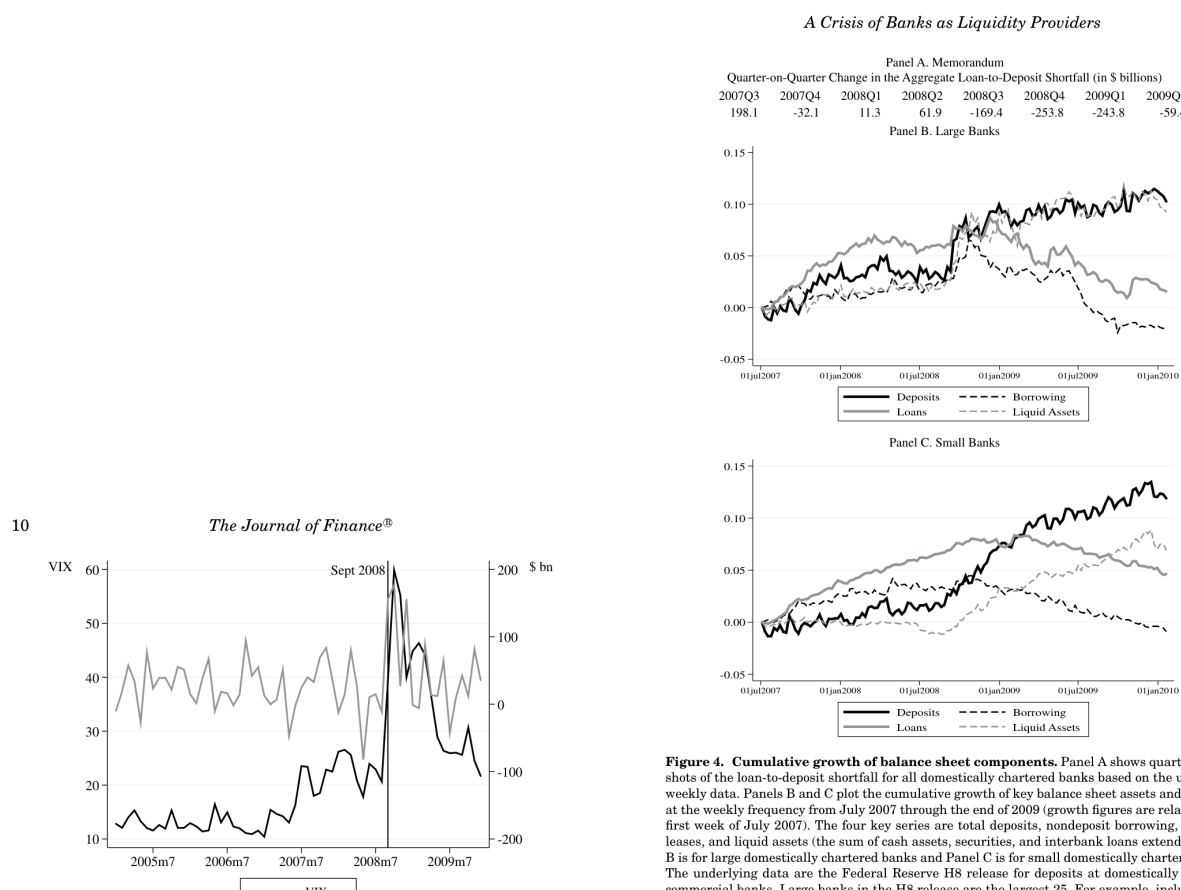
7.2 Figures 3–4: Market stress, deposits, and balance-sheet shortfalls

Read:

- Figure 3 compares the VIX with net deposit flows into commercial banks.
- Before Lehman, market stress rose without a strong bank-deposit inflow.
- Figure 4 shows balance-sheet growth for large and small banks: deposits, borrowing, loans, and liquid assets.
- The loan-to-deposit shortfall widened in the first crisis phase and closed only after large deposit inflows and public guarantees.

Detailed interpretation: Figure 3 is the aggregate timing evidence: market stress does not automatically mean deposits flood into banks. Figure 4 connects the flow problem to bank balance sheets: lending grew, commitments were drawn, and deposits lagged.

Economic message: The bank liquidity-provision model works only if deposit funding arrives when commitments are drawn. Figures 3 and 4 show that this hedge failed early in the 2007–2009 crisis.



(a) Figure 3: Market stress and net flows into deposits

(b) Figure 4: Cumulative growth of balance-sheet components

7.3 Tables I–II: Aggregate deposit growth and lending changes

Read:

- Table I shows weak core-deposit growth before 2008Q3 and a sharp deposit inflow after Lehman-related policy support.
- Table II compares average weekly deposit and lending changes during the recent crisis and the 1998 LTCM crisis.
- The 1998 crisis shows the classic flight to bank deposits; the first phase of 2007–2009 does not.

Detailed interpretation: These tables provide the aggregate benchmark. The banking system did not receive the same stabilizing deposit inflow at the start of this crisis as in previous market-stress episodes. The reversal after Lehman is consistent with the importance of stronger public guarantees.

Economic message: The banking system’s aggregate backup-liquidity function depends on confidence in bank liabilities and public insurance.

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Table I
Cumulative Deposit Growth during the 2007 to 2009
Financial Crisis (in %)

This table presents an extract of cumulative growth statistics for certain deposit accounts based on the quarterly Call Reports over the period from 2007Q3 to 2009Q1 (see the online version of this article for full balance sheet details). Panel A shows the growth of insured deposits, core deposits, and large time deposits over the period relative to initial 2007Q2 levels for large banks. Similarly, Panel B shows the deposit growth figures for small banks, and the bottom panel summarizes overall deposit inflows into the banking system in billions of dollars over the period 2007Q4 to 2009Q1. These aggregate flow of funds match the data reported in He, Khang, and Krishnamurthy (2010) for the corresponding period (where core deposits are reported to have increased by roughly \$800 billion while large time deposits decreased by \$200 billion).

	2007Q3	2007Q4	2008Q1	2008Q2	2008Q3	2008Q4	2009Q1
Panel A: Large Banks (Largest 25 Banks, H8 Criteria)							
Insured deposits	0.1	0.8	1.7	1.7	3.9	4.6	5.6
Core deposits	0.2	2.4	3.8	3.4	6.3	9.7	10.8
Large time deposits	0.6	1.5	1.8	1.3	2.8	2.0	1.2
Panel B: Small Banks							
Insured deposits	0.0	0.0	0.7	1.0	2.3	3.0	2.0

(a) Table I: Consolidated deposit growth

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Table II
Average Weekly Deposit Funding and Lending Changes (in \$ Billions)

This table reports the average weekly deposit funding and lending changes that occurred in the 2007 to 2009 crisis and compares these changes to those that occurred during the 1998 LTCM Crisis. The sample is weekly from the Federal Reserve H8 Release for all reporting domestically chartered banks.

	1998 LTCM Crisis		2007 to 2009 Financial Crisis			
	Previous Five Years to Crisis	Fall 1998	Previous Five Years to Crisis	July 4, 2007, to Sept 10, 2008 (Pre-Lehman Failure)	Period Immediately Following Lehman Failure	Oct 8, 2008, to July 1, 2009
Deposits	2.1	8.7	6.6	6.5	94.7	9.0
of which, core deposits	1.4	7.7	5.2	4.7	59.2	12.2
Loans	3.5	8.1	8.1	9.5	38.2	−5.3
of which, C&I loans	1.0	2.1	0.9	3.1	15.8	−2.5

(b) Table II: Weekly deposit funding and lending changes

7.4 Tables III–IV: Summary statistics and pre/post crisis behavior

Read:

- Table III reports summary statistics for bank exposures, funding, assets, and correlations.
- Table IV shows that deposit rates and balance-sheet behavior change in crisis periods.
- The sample has meaningful cross-sectional variation in unused commitments, wholesale funding, NPLs, size, capital, and real-estate exposure.

Detailed interpretation: Table III establishes that liquidity demand risk is not identical to solvency risk. Table IV begins the bank-level evidence by showing how balance-sheet behavior shifts around the crisis.

Economic message: Cross-bank variation in unused commitments makes it possible to test which banks were most liquidity-short when the aggregate funding hedge failed.

Table III
Summary Statistics for Regression Bank Panel, 1990Q1 to 2009Q4

This table presents summary statistics for the main regression variables. These statistics are calculated over the regression sample (and thus exclude mergers, non-U.S.-domiciled banking organizations, and those below \$100 million in total assets). See the Internet Appendix for variable definitions and sources.

	Mean	Standard Deviation	25 th Percentile	Median	75 th Percentile	Obs.
Dependent variables (Call Reports)						
Interest rate, large time deposits (implicit, % annual)	4.526	1.571	3.344	4.567	5.538	238,557
Interest rate, core deposits (implicit, % annual)	3.406	1.443	2.265	3.397	4.295	238,543
Quarterly growth of deposits	0.011	0.031	-0.006	0.008	0.028	253,492
Quarterly growth of core deposits	0.008	0.028	-0.006	0.006	0.023	253,492
Quarterly growth of brokered deposits	0.001	0.008	0.000	0.000	0.000	236,763
Transaction account guarantee program deposit share (2008Q4)	0.033	0.045	0.004	0.021	0.044	21,674
Quarterly growth of loans	0.010	0.029	-0.003	0.010	0.025	253,492
Quarterly growth of C&I loans	0.001	0.013	-0.003	0.001	0.006	253,492
Quarterly growth of credit (loans + commitments)	0.011	0.032	-0.004	0.010	0.027	253,492
Loan-deposit shortfall	-0.154	0.240	-0.290	-0.176	-0.057	253,492
Quarterly growth of loan-deposit shortfall	0.000	0.038	-0.021	0.000	0.021	253,492
Quarterly growth of federal funds purchased and repo borrowing	0.000	0.013	0.000	0.000	0.001	253,492
Quarterly growth of other borrowed money (total)	0.001	0.016	0.000	0.000	0.001	210,745
Dependent variables (Bank Rate Monitor 1997 to 2009)						
Interest checking rate (%)	0.610	0.601	0.150	0.400	1.000	107,466
CD 12-month rate (%)	3.170	1.540	1.740	3.200	4.500	107,298
CD 24-month rate (%)	3.448	1.412	2.230	3.444	4.640	104,874
CD 60-month rate (%)	4.023	1.153	3.200	4.020	4.890	97,235
Covariates						
Unused commitment ratio	0.127	0.087	0.070	0.117	0.170	257,078
Liquidity ratio (excludes MBS/ABS)	0.242	0.144	0.141	0.218	0.312	215,100
Wholesale funding ratio	0.198	0.144	0.103	0.171	0.255	258,108
Net wholesale funding ratio (wholesale - liquid)	-0.030	0.219	-0.163	-0.031	0.099	215,100
Nonperforming loans to loans	0.012	0.020	0.002	0.007	0.014	228,042
Capital ratio (book capital to	0.091	0.045	0.076	0.089	0.106	258,108

(a) Table III: Summary statistics, part 1

Table III—Continued

Pairwise Correlation Coefficients for Key Covariates in Regression Bank Panel 1990Q1 to 2009Q4						
	1	2	3	4	5	6
1 Unused commitment ratio	1.000					
2 Net wholesale funding ratio	0.053	1.000				
3 Nonperforming loans to loans	-0.036	0.044	1.000			
4 Capital ratio (book capital to assets)	0.059	-0.214	0.053	1.000		
5 Indicator for large banks	0.225	0.087	0.009	0.003	1.000	
6 Real estate loan share	0.045	0.007	0.074	0.272	-0.086	1.000

(b) Table III: Summary statistics, part 2

Table IV
The Relationship between the Deposit Interest Rate and Liquidity Demand Risk in the Crisis

This table reports regressions relating the rate paid on deposits to a bank's liquidity demand risk during the crisis. The regressions allow for two phases of the crisis: 2007Q3 to 2008Q2 (*Crisis1*) and 2008Q3 to 2009Q2 (*Crisis2*). The dependent variable is the implicit rate on deposits, % annual (Call Reports). All specifications are panel regressions over the period 1994 to 2009 with fixed effects for bank organizations and quarterly time dummies (unless otherwise noted). The reported R^2 is within- R^2 . All regressions control for district time trends and for the deposit-weighted geographic market deposit concentration. The standard errors used in calculating significance levels are clustered at the bank organization level and are shown in parentheses. See the Internet Appendix for variable definitions and details about the bank panel. ***, **, and * indicate 1%, 5%, and 10% significance, respectively.

	(1) Large Time	(2) Large Time	(3) Core	(4) Core
Unused Commitment Ratio _{<i>t-1</i>}	0.139 (0.095)	-0.810*** (0.106)	-0.329*** (0.093)	-1.804*** (0.117)
Unused Commitment Ratio _{<i>t-1</i>} × Crisis1	0.592*** (0.119)	0.710*** (0.139)	0.033 (0.095)	0.725*** (0.217)
Unused Commitment Ratio _{<i>t-1</i>} × Crisis2	-0.577*** (0.131)	-0.810*** (0.156)	-0.590*** (0.102)	-0.617*** (0.176)
Controls				
Net Wholesale Funding _{<i>t-1</i>}	0.310*** (0.043)	0.529*** (0.036)	0.012 (0.037)	0.890*** (0.044)
Net Wholesale Funding _{<i>t-1</i>} × Crisis1	0.119*** (0.046)	0.162*** (0.053)	0.467*** (0.042)	0.575*** (0.064)
Net Wholesale Funding _{<i>t-1</i>} × Crisis2	0.019 (0.051)	0.066 (0.058)	0.248*** (0.038)	0.312*** (0.059)
NPL to Loans _{<i>t-1</i>}	1.073*** (0.236)	0.687** (0.316)	0.218 (0.170)	1.122*** (0.346)
NPL to Loans _{<i>t-1</i>} × Crisis1	-0.549 (0.465)	0.302 (0.521)	0.784*** (0.269)	2.302** (0.952)
NPL to Loans _{<i>t-1</i>} × Crisis2	1.083*** (0.316)	1.872*** (0.395)	1.050*** (0.209)	2.373*** (0.457)
Capital Ratio _{<i>t-1</i>}	-0.834** (0.328)	-0.098 (0.223)	-1.834*** (0.260)	0.381 (0.259)
Capital Ratio _{<i>t-1</i>} × Crisis1	0.775*** (0.220)	0.660*** (0.233)	0.389 (0.306)	0.176 (0.417)
Capital Ratio _{<i>t-1</i>} × Crisis2	-0.418 (0.317)	-0.804** (0.354)	0.394* (0.218)	-0.542* (0.333)
Large Bank Indicator	-0.244* (0.146)	-0.204 (0.131)	-0.038 (0.122)	-0.393*** (0.087)
Large Bank Indicator × Crisis1	-0.167 (0.138)	-0.215 (0.163)	-0.307*** (0.073)	-0.664*** (0.093)
Large Bank Indicator × Crisis2	-0.142 (0.103)	-0.133 (0.143)	-0.002 (0.070)	-0.143 (0.094)
Real Estate Loan Share _{<i>t-1</i>}	0.025 (0.060)	0.117*** (0.042)	-0.044 (0.054)	-0.021 (0.050)

Figure 5: Table IV: Behavior before and after crisis

7.5 Tables V–VI: Deposit inflows, lending, and liquidity demand risk

Read:

- Table V shows that high-commitment banks experienced weaker deposit growth in the first crisis phase, even while offering higher rates.
- Table VI shows that high-commitment banks increased lending due to drawdowns but reduced new credit and developed larger loan-to-deposit shortfalls.
- The effects differ across Crisis1 and Crisis2, consistent with the post-Lehman funding reversal.

Detailed interpretation: These are the core micro tables. A high-commitment bank is precisely the bank expected to provide liquidity to firms. During Crisis1, such banks could not attract enough deposits and therefore faced a growing gap between lending demand and deposit funding.

Economic message: Credit lines were honored, but the private bank-deposit system did not fully fund them. Liquidity provision became dependent on other adjustment margins.

Table V
The Relationship between Deposit Inflows and Liquidity Demand Risk in the Crisis
This table reports regressions relating the growth of different categories of deposits to a bank's liquidity demand risk during the crisis. The regressions allow for two phases of the crisis: 2007Q3 to 2008Q2 (*Crisis1*) and 2008Q3 to 2009Q2 (*Crisis2*). Specifications (1) to (4) are panel regressions over the period 1994 to 2009 with fixed effects for bank organizations and quarterly time dummies. The reported R^2 is within- R^2 . The specification in column (5) is cross-sectional for 2008Q4 when the Transaction Account Guarantee (TAG) was introduced. Column (5) also controls for the share of transaction deposits. Regressions control for district time trends and for the deposit-weighted geographic market deposit concentration. The standard errors used in calculating significance levels are clustered at the bank organization level and are shown in parentheses. See the Internet Appendix for variable definitions and details about the bank panel. ***, **, and * indicate 1%, 5%, and 10% significance, respectively.

	(1)	(2)	(3)	(4)	(5)
	Δ Deposits/ Assets ₋₁	Δ Core Deposits/ Assets ₋₁	Δ Insured Deposits/ Assets ₋₁	Δ Brokered Deposits/ Assets ₋₁	TAG Deposits (2008Q4)/ Assets ₋₁
Unused Commitment Ratio ₋₁	0.034*** (0.004)	0.025*** (0.003)	0.014*** (0.002)	0.008*** (0.001)	0.174*** (0.016)
Unused Commitment Ratio ₋₁ $\times$ Crisis1	-0.018*** (0.006)	-0.017*** (0.005)	0.007 (0.004)	0.007*** (0.002)	
Unused Commitment Ratio ₋₁ $\times$ Crisis2	0.018*** (0.006)	0.005 (0.005)	0.030*** (0.004)	0.018*** (0.002)	
Controls					
Net Wholesale Funding ₋₁	0.008*** (0.002)	0.006*** (0.002)	0.028*** (0.001)	0.008*** (0.000)	-0.023*** (0.007)
Net Wholesale Funding ₋₁ $\times$ Crisis1	-0.008*** (0.002)	0.004*** (0.002)	0.007*** (0.002)	0.002*** (0.001)	
Net Wholesale Funding ₋₁ $\times$ Crisis2	0.003*** (0.002)	0.010*** (0.002)	0.007*** (0.002)	0.006*** (0.001)	
NPL to Loans ₋₁	-0.192*** (0.014)	-0.139*** (0.010)	-0.100*** (0.008)	-0.040*** (0.004)	0.073* (0.040)
NPL to Loans ₋₁ $\times$ Crisis1	0.009 (0.015)	0.025 (0.015)	0.041*** (0.013)	0.014 (0.009)	
NPL to Loans ₋₁ $\times$ Crisis2	0.058*** (0.019)	0.042*** (0.015)	0.056*** (0.013)	0.002 (0.006)	

(Continued)

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(a) Table V: Deposit inflows and liquidity demand risk, part 1

Table V—Continued

	(1)	(2)	(3)	(4)	(5)
	Δ Deposits/ Assets ₋₁	Δ Core Deposits/ Assets ₋₁	Δ Insured Deposits/ Assets ₋₁	Δ Brokered Deposits/ Assets ₋₁	TAG Deposits (2008Q4)/ Assets ₋₁
Capital Ratio ₋₁	0.189*** (0.012)	0.158*** (0.009)	0.058*** (0.007)	0.018*** (0.001)	-0.031 (0.029)
Capital Ratio ₋₁ $\times$ Crisis1	0.024* (0.013)	0.023** (0.011)	0.040*** (0.008)	0.007** (0.003)	
Capital Ratio ₋₁ $\times$ Crisis2	0.072*** (0.014)	0.052*** (0.012)	0.052*** (0.010)	0.022*** (0.004)	
Large Bank Indicator	-0.068** (0.003)	-0.005** (0.002)	-0.002 (0.002)	-0.002*** (0.001)	0.013 (0.006)
Large Bank Indicator $\times$ Crisis1					
Large Bank Indicator $\times$ Crisis2					
Real Estate Loan Share ₋₁	-0.005 (0.003)	-0.005 (0.003)	0.009*** (0.002)	-0.005*** (0.001)	
Real Estate Loan Share ₋₁ $\times$ Crisis1	-0.012*** (0.002)	-0.012*** (0.002)	-0.006*** (0.001)	-0.001 (0.006)	-0.032*** (0.005)
Real Estate Loan Share ₋₁ $\times$ Crisis2	-0.026*** (0.002)	-0.026*** (0.002)	-0.011*** (0.001)	0.002** (0.001)	
Bank Fixed Effects	Yes	Yes	Yes	Yes	No
Observations	197,198	197,198	197,198	196,655	3,522
R^2	0.07	0.08	0.32	0.04	0.14

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(b) Table V: Deposit inflows and liquidity demand risk, part 2

Table VI
The Relationship between Lending, Loan-to-Deposit Shortfalls, and Liquidity Demand Risk in the Crisis
This table reports regressions relating lending growth and loan-to-deposit shortfalls to a bank's liquidity demand risk during the crisis. The regressions allow for two phases of the crisis: 2007Q3 to 2008Q2 (*Crisis1*) and 2008Q3 to 2009Q2 (*Crisis2*). All specifications are panel regressions over the period 1994 to 2009 with fixed effects for bank organizations and quarterly time dummies. The reported R^2 is within- R^2 . Regressions control for district time trends and for the deposit-weighted geographic market deposit concentration. The standard errors used in calculating significance levels are clustered at the bank organization level and are shown in parentheses. See the Internet Appendix for variable definitions and details about the bank panel. ***, **, and * indicate 1%, 5%, and 10% significance, respectively.

	(1)	(2)	(3)	(4)	(5)
	Δ Loans/ Assets ₋₁	Δ C&I Loans/ Assets ₋₁	Δ Loans + Commitments/ Assets ₋₁	Δ Loans - Deposits/ Assets ₋₁	Δ Loans - Deposits/ Assets ₋₁
Unused Commitment Ratio ₋₁	0.134*** (0.006)	0.018*** (0.001)	-0.016*** (0.004)	-0.042*** (0.017)	0.099*** (0.006)
Unused Commitment Ratio ₋₁ $\times$ Crisis1	0.014*** (0.005)	0.004*** (0.002)	-0.023*** (0.005)	0.118*** (0.019)	0.028*** (0.007)
Unused Commitment Ratio ₋₁ $\times$ Crisis2	-0.002 (0.006)	-0.002 (0.002)	-0.052*** (0.006)	0.065*** (0.017)	-0.021*** (0.007)
Controls					
Net Wholesale Funding ₋₁	-0.012*** (0.001)	-0.003*** (0.000)	-0.029*** (0.001)	0.513*** (0.008)	-0.084*** (0.002)
Net Wholesale Funding ₋₁ $\times$ Crisis1	-0.005*** (0.002)	-0.001*** (0.002)	-0.010*** (0.006)	0.027*** (0.002)	0.001 (0.002)
Net Wholesale Funding ₋₁ $\times$ Crisis2	-0.018*** (0.002)	-0.004*** (0.001)	-0.027*** (0.002)	0.014* (0.006)	-0.022*** (0.003)
NPL to Loans ₋₁	-0.303*** (0.017)	-0.054*** (0.005)	-0.381*** (0.020)	-0.254*** (0.035)	-0.198*** (0.012)
NPL to Loans ₋₁ $\times$ Crisis1	-0.089*** (0.017)	0.005 (0.006)	-0.067*** (0.051)	-0.078 (0.015)	-0.077*** (0.015)
NPL to Loans ₋₁ $\times$ Crisis2	0.018 (0.019)	0.018*** (0.005)	-0.014 (0.020)	-0.070* (0.030)	-0.046*** (0.016)

(Continued)

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(a) Table VI: Lending and liquidity demand risk, part 1

Table VI—Continued

	(1)	(2)	(3)	(4)	(5)
	Δ Loans/ Assets ₋₁	Δ C&I Loans/ Assets ₋₁	Δ Loans + Commitments/ Assets ₋₁	Δ Loans - Deposits/ Assets ₋₁	Δ Loans - Deposits/ Assets ₋₁
Capital Ratio ₋₁	0.015 (0.010)	0.004 (0.004)	0.022** (0.011)	0.725*** (0.053)	-0.176*** (0.014)
Capital Ratio ₋₁ $\times$ Crisis1	0.042*** (0.010)	0.005 (0.004)	0.009*** (0.012)	0.045 (0.038)	0.027 (0.017)
Capital Ratio ₋₁ $\times$ Crisis2	0.054*** (0.011)	0.010*** (0.004)	0.010*** (0.012)	0.010 (0.050)	-0.021 (0.017)
Large Bank Indicator	0.008*** (0.003)	0.005*** (0.001)	0.010*** (0.002)	-0.054*** (0.015)	0.003 (0.004)
Large Bank Indicator $\times$ Crisis1					
Large Bank Indicator $\times$ Crisis2					
Real Estate Loan Share ₋₁	0.001 (0.003)	-0.001 (0.001)	-0.002 (0.002)	-0.006 (0.008)	0.007* (0.004)
Real Estate Loan Share ₋₁ $\times$ Crisis1					
Real Estate Loan Share ₋₁ $\times$ Crisis2					
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes
Observations	197,198	197,198	197,198	197,198	197,198
R^2	0.15	0.03	0.11	0.50	0.10

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(b) Table VI: Lending and liquidity demand risk, part 2

7.6 Tables VII–VIII: Adjustment margins and solvency splits

Read:

- Table VII shows how high-commitment banks adjusted by using liquid assets and borrowing, especially other borrowed money such as FHLB advances.
- Table VIII partitions banks by solvency problems measured by nonperforming loans, real-estate loan exposure, and capital.
- The commitment channel remains relevant after distinguishing liquidity risk from solvency risk.

Detailed interpretation: Table VII explains how liquidity provision continued despite insufficient deposits: banks used buffers and public / quasi-public funding sources. Table VIII then addresses an important alternative story: the effects are not simply because high-commitment banks were insolvent banks.

Economic message: The crisis was not only a solvency crisis. It was also a liquidity-demand crisis on both sides of bank balance sheets.

Table VII
The Relationship between Liquid Assets, Borrowed Money, and Liquidity Demand Risk in the Crisis
 This table reports regressions relating the growth of liquid assets and nondeposit borrowing to a bank's liquidity demand risk during the crisis. The regressions allow for two phases of the crisis: 2007Q3 to 2008Q2 (Crisis1) and 2008Q3 to 2009Q2 (Crisis2). The dependent variable is column (3), "other borrowed money" (BCFD3100), is the sum of borrowing from the FHLBs (column (4) and "other" (includes borrowing from the Federal Reserve) (column (5)), which are available from 2001 only (see Call Report schedule BC MI). All specifications are panel regressions over the period 1994 to 2009 (2001 to 2009 in column (4) and (5)) with fixed effects for bank organizations and quarterly time dummies. The reported R^2 is within R^2 . Regressions control for district time trends and for the deposit-weighted geographic market deposit concentration. The standard errors used in calculating significance levels are clustered at the bank organization level and are shown in parentheses. See the Internet Appendix for variable definitions and details about the bank panel. ***, **, and * indicate 1%, 5%, and 10% significance, respectively.

	(1) Δ (Liquid Assets)/ Assets, $_{t-1}$	(2) Δ (Federal Funds and Repo Borrowing)/ Assets, $_{t-1}$	(3) Δ (Other Borrowed Money Total)/ Assets, $_{t-1}$	(4) Δ (Other Borrowed Money FHLB)/ Assets, $_{t-1}$	(5) Δ (Other Borrowed Money Other)/ Assets, $_{t-1}$
Unused Commitment Ratio, $_{t-1}$	-0.073*** (0.005)	0.007*** (0.001)	0.012*** (0.002)	0.016*** (0.000)	0.001 (0.000)
Unused Commitment Ratio, $_{t-1} \times$ Crisis1	-0.011** (0.006)	0.004*** (0.002)	0.013*** (0.002)	0.011*** (0.001)	-0.001 (0.003)
Unused Commitment Ratio, $_{t-1} \times$ Crisis2	0.009 (0.006)	-0.009*** (0.002)	-0.003 (0.001)	-0.004* (0.002)	0.002** (0.001)
Controls					
Net Wholesale Funding, $_{t-1}$	0.067*** (0.002)	-0.010*** (0.000)	-0.010*** (0.001)	-0.013*** (0.001)	-0.001*** (0.000)
Net Wholesale Funding, $_{t-1} \times$ Crisis1	0.002 (0.002)	0.000 (0.001)	0.005*** (0.001)	0.004*** (0.001)	0.001** (0.001)
Net Wholesale Funding, $_{t-1} \times$ Crisis2	0.008*** (0.002)	-0.003*** (0.001)	-0.011*** (0.001)	-0.010*** (0.001)	0.000 (0.000)
NPL to Loans, $_{t-1}$	0.009 (0.002)	-0.004*** (0.001)	-0.010*** (0.001)	-0.012*** (0.001)	-0.001 (0.001)
NPL to Loans, $_{t-1} \times$ Crisis1	0.034* (0.018)	-0.012*** (0.004)	-0.020*** (0.008)	-0.024*** (0.009)	-0.001 (0.001)
NPL to Loans, $_{t-1} \times$ Crisis2	0.007 (0.016)	0.000 (0.003)	0.009* (0.005)	0.007 (0.005)	0.000 (0.001)

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(Continued)

(a) Table VII: Liquid assets and borrowing, part 1

	(1) Δ (Liquid Assets)/ Assets, $_{t-1}$	(2) Δ (Federal Funds and Repo Borrowing)/ Assets, $_{t-1}$	(3) Δ (Other Borrowed Money Total)/ Assets, $_{t-1}$	(4) Δ (Other Borrowed Money FHLB)/ Assets, $_{t-1}$	(5) Δ (Other Borrowed Money Other)/ Assets, $_{t-1}$
Capital Ratio, $_{t-1}$	0.177*** (0.013)	0.006*** (0.002)	0.020*** (0.004)	0.024*** (0.004)	0.001 (0.001)
Capital Ratio, $_{t-1} \times$ Crisis1	-0.021 (0.014)	-0.001 (0.002)	0.007 (0.007)	0.001 (0.004)	0.003 (0.002)
Capital Ratio, $_{t-1} \times$ Crisis2	0.015 (0.015)	0.005* (0.003)	0.005 (0.005)	0.004 (0.004)	0.002 (0.002)
Large Bank Indicator	-0.007*** (0.003)	-0.001 (0.001)	0.000 (0.001)	0.002 (0.002)	-0.001 (0.001)
Large Bank Indicator $\times$ Crisis1	0.001 (0.003)	0.000 (0.001)	0.003 (0.001)	0.001 (0.002)	0.003** (0.001)
Large Bank Indicator $\times$ Crisis2	-0.016*** (0.005)	0.000 (0.001)	0.000 (0.001)	0.005*** (0.001)	-0.002 (0.001)
Real Estate Loan Share, $_{t-1}$	0.005 (0.012)**	0.002 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)
Real Estate Loan Share, $_{t-1} \times$ Crisis1	-0.022*** (0.002)	0.001* (0.000)	0.005*** (0.001)	0.008*** (0.001)	0.000 (0.002)
Real Estate Loan Share, $_{t-1} \times$ Crisis2	0.002 (0.002)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.002 (0.001)
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes
Observations	197,198	197,198	197,198	122,324	122,324
R^2	0.07	0.02	0.03	0.03	0.01

(b) Table VII: Liquid assets and borrowing, part 2

Table VIII
The Relationship between Solvency Risk and Liquidity Demand Risk in the Crisis: Are Banks with Solvency Problems More Vulnerable to Liquidity Demand Risk?
 This table reports regressions testing whether banks with weaker fundamentals were more vulnerable to the onset of the crisis. The regressions allow for two phases of the crisis: 2007Q3 to 2008Q2 (Crisis1) and 2008Q3 to 2009Q2 (Crisis2). The sample is partitioned into high- and low-solvency problems based on nonperforming loans (columns (1) and (2)), real estate loans (columns (3) and (4)), or capital (columns (5) and (6)). "High" solvency risk is proxied by the set of banks with an above-median nonperforming loan ratio, above-median real estate loan share, or below-median capital ratio (Table III). All specifications are panel regressions over the period 1994 to 2009 with fixed effects for bank organizations and quarterly time dummies. The reported R^2 is within R^2 . Regressions control for district time trends and for the deposit-weighted geographic market deposit concentration. The standard errors used in calculating significance levels are clustered at the bank organization level and are shown in parentheses. See the Internet Appendix for variable definitions and details about the bank panel. ***, **, and * indicate 1%, 5%, and 10% significance, respectively.

	(1) Nonperforming Loans High	(2) Nonperforming Loans Low	(3) Real Estate Loan Share High	(4) Real Estate Loan Share Low	(5) Capital Ratio High	(6) Capital Ratio Low
Unused Commitment Ratio, $_{t-1}$	0.046 (0.129)	0.369*** (0.127)	0.219* (0.114)	0.087 (0.150)	0.173 (0.148)	0.043 (0.124)
Unused Commitment Ratio, $_{t-1} \times$ Crisis1	0.088*** (0.150)	0.450*** (0.170)	0.451*** (0.140)	0.100* (0.184)	0.577*** (0.191)	0.530*** (0.147)
Unused Commitment Ratio, $_{t-1} \times$ Crisis2	-0.351** (0.150)	-0.615*** (0.222)	-0.548*** (0.158)	-0.631*** (0.312)	-0.153 (0.209)	-0.826*** (0.159)
Other bank controls included	Yes	Yes	Yes	Yes	Yes	Yes
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	95,731	96,393	111,204	84,920	84,466	111,658
R^2	0.73	0.74	0.72	0.72	0.71	0.73

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(Continued)

(a) Table VIII: Solvency and liquidity risk, part 1

	(7) Nonperforming Loans High	(8) Nonperforming Loans Low	(9) Real Estate Loan Share High	(10) Real Estate Loan Share Low	(11) Capital Ratio High	(12) Capital Ratio Low
Unused Commitment Ratio, $_{t-1}$	0.029*** (0.001)	0.027*** (0.001)	0.063*** (0.005)	0.069* (0.005)	0.036*** (0.005)	0.028*** (0.005)
Unused Commitment Ratio, $_{t-1} \times$ Crisis1	-0.016** (0.007)	-0.013 (0.009)	-0.028*** (0.006)	-0.001 (0.010)	-0.015 (0.009)	-0.015** (0.008)
Unused Commitment Ratio, $_{t-1} \times$ Crisis2	0.019** (0.008)	0.011 (0.010)	0.020*** (0.007)	0.017* (0.009)	0.021* (0.011)	0.022*** (0.007)
Other bank controls included	Yes	Yes	Yes	Yes	Yes	Yes
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	100,013	97,185	111,229	85,869	85,178	112,020
R^2	0.08	0.07	0.08	0.08	0.09	0.07

(b) Table VIII: Solvency and liquidity risk, part 2

7.7 Tables IX–X: Aggregate funding shift and effective drawdowns

Read:

- Table IX splits Crisis2 into 2008Q3 and 2008Q4–2009Q2 to isolate the post-Lehman funding reversal.
- Table X compares banks with above- and below-median effective commitment drawdowns.
- Effective drawdown banks show sharper liquidity stress, stronger deposit-rate responses, and larger liquid-asset adjustments.

Detailed interpretation: Table IX clarifies that the post-Lehman period was not a continuation of the initial funding problem. Public support changed the funding environment. Table X shows that realized commitment usage matters, not just potential exposure.

Economic message: The paper's mechanism is strongest where commitments actually turn into funded loans.

Table IX
The Aggregate Shift in Funding: Separating Crisis2 into 2008Q3 (Post-Lehman) and 2008Q4 to 2009Q2

This table reports regressions relating the rate paid on deposits, changes in funding, and lending to a bank's liquidity demand risk during the crisis. The regressions allow for three phases of the crisis: 2007Q3 to 2008Q2 (*Crisis1*), 2008Q3 (*Crisis2 08Q3*), and 2008Q4 to 2009Q2 (*Crisis2 08Q4-09Q2*). All specifications are panel regressions over the period 1994 to 2009 with fixed effects for bank organizations and quarterly time dummies. The reported R^2 is within- R^2 . Regressions control for district time trends and for the deposit-weighted geographic market deposit concentration. The standard errors used in calculating significance levels are clustered at the bank organization level and are shown in parentheses. See the Internet Appendix for variable definitions, details about the bank panel, and for full controls. *** and * indicate 1% and 10% significance, respectively

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Rate on Large Time Deposits	Rate on Core Deposits	Δ Deposits/ Assets _{t-1}	Δ (Loans – Deposits)/ Assets _{t-1}	Δ (Liquid Assets)/ Assets _{t-1}	Δ (Federal Funds and Repo Borrowing)/ Assets _{t-1}	Δ (Other Borrowed Money Total)/ Assets _{t-1}
Unused Commitment Ratio _{t-1}	0.140 (0.095)	-0.329*** (0.093)	0.034*** (0.004)	0.099*** (0.006)	-0.073*** (0.005)	0.008*** (0.001)	0.012*** (0.002)
Unused Commitment Ratio _{t-1} × Crisis1	0.594*** (0.120)	0.033 (0.095)	-0.016*** (0.006)	0.028*** (0.007)	-0.011* (0.006)	0.004*** (0.002)	0.013*** (0.002)
Unused Commitment Ratio _{t-1} × Crisis2 08Q3	-0.315* (0.174)	-0.541*** (0.109)	0.049*** (0.010)	-0.058*** (0.012)	0.030*** (0.010)	-0.015*** (0.004)	-0.007 (0.005)
Unused Commitment Ratio _{t-1} × Crisis2 08Q4-09Q2	-0.656*** (0.145)	-0.605*** (0.112)	0.008 (0.007)	-0.007 (0.008)	0.002 (0.007)	-0.007*** (0.002)	-0.002 (0.003)
Other bank controls included	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	196,124	196,151	197,198	197,198	197,198	197,198	197,198
R^2	0.74	0.89	0.08	0.10	0.07	0.02	0.03

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Figure 10: Table IX: Aggregate shift in funding

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Table X
Partitioning Banks by Their Unused Commitments Constraint: Effective Drawdowns

This table reports regressions testing the hypothesis that the crisis should have especially constrained banks with greater effective, net potential, drawdowns. The regressions allow for two phases of the crisis: 2007Q3 to 2008Q2 (*Crisis1*) and 2008Q3 to 2009Q2 (*Crisis2*). Note that “High” drawdowns is proxied by the set of banks with more than four quarters during the eight-quarter crisis of negative growth of their available commitments. These banks also roughly correspond to banks with above-median decline in unused commitments. All specifications are panel regressions over the period 1994 to 2009 with fixed effects for bank organizations and quarterly time dummies. The reported R^2 is within- R^2 . Regressions control for district time trends and for the deposit-weighted geographic market deposit concentration. The standard errors used in calculating significance levels are clustered at the bank organization level and are shown in parentheses. See the Internet Appendix for variable definitions, details about the bank panel, and the full details for the controls. *** and ** indicate 1% and 5% significance, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	Rate on Large Time Deposits High	Rate on Large Time Deposits Low	Δ Deposits/ Assets _{t-1} High	Δ Deposits/ Assets _{t-1} Low	Δ Loans/ Assets _{t-1} High	Δ Loans/ Assets _{t-1} Low
Unused Commitment Ratio _{t-1}	0.131 (0.141)	0.090 (0.148)	0.048*** (0.006)	0.022*** (0.003)	0.146*** (0.011)	0.113*** (0.009)
Unused Commitment Ratio _{t-1} × Crisis1	0.748*** (0.160)	0.390*** (0.175)	-0.021*** (0.007)	-0.009 (0.009)	0.030*** (0.007)	0.008 (0.006)
Unused Commitment Ratio _{t-1} × Crisis2	-0.552*** (0.212)	-0.619*** (0.194)	0.015 (0.010)	0.019*** (0.007)	0.007 (0.008)	-0.007 (0.008)
Other bank controls included	Yes	Yes	Yes	Yes	Yes	Yes
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	66,511	84,442	68,645	84,105	68,645	84,105
R^2	0.78	0.75	0.09	0.07	0.22	0.12

(Continued)

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Table X—Continued

	(7)	(8)	(9)	(10)	(11)	(12)
	Δ Brokered Deposits/ Assets _{t-1} High	Δ Brokered Deposits/ Assets _{t-1} Low	Δ Other Borrowed Money Total/ Assets _{t-1} High	Δ Other Borrowed Money Total/ Assets _{t-1} Low	Δ (Liquid Assets)/ Assets _{t-1} High	Δ (Liquid Assets)/ Assets _{t-1} Low
Unused Commitment Ratio _{t-1}	0.011*** (0.002)	0.006*** (0.001)	0.011*** (0.002)	0.011*** (0.003)	-0.073*** (0.008)	-0.068*** (0.007)
Unused Commitment Ratio _{t-1} × Crisis1	0.010*** (0.003)	0.004 (0.004)	0.017*** (0.004)	0.009*** (0.003)	-0.027*** (0.006)	-0.001 (0.009)
Unused Commitment Ratio _{t-1} × Crisis2	0.002*** (0.003)	0.011*** (0.003)	-0.001 (0.004)	-0.003 (0.003)	-0.001 (0.009)	0.016** (0.008)
Other bank controls included	Yes	Yes	Yes	Yes	Yes	Yes
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	68,567	83,787	68,645	84,105	68,645	84,105
R^2	0.06	0.03	0.04	0.03	0.07	0.08

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(a) Table X: Effective drawdowns, part 1

(b) Table X: Effective drawdowns, part 2

7.8 Tables XI–XII: Previous crises versus the 2007–2009 crisis

Read:

- Table XI revisits the relationship between market stress, commitments, deposits, and nondeposit liabilities.
- Table XII does the same for loans and total credit.
- In previous crises, unused commitments interacted positively with market stress; in 2007–2009, the triple interaction indicates the relationship was different and weaker or reversed.

Detailed interpretation: These tables are the historical comparison. They show that banks did act as liquidity providers in earlier market-stress episodes, but the 2007–2009 crisis disrupted that pattern because banks themselves were under stress.

Economic message: Liquidity-provision capacity is state-dependent. The bank-centered nature of the crisis determines whether banks can hedge market liquidity demand.

Table XI
The Relationship between Market Stress and Growth of Deposits and Nondeposit Liabilities: Was It Different in the 2007 to 2009 Crisis? Revisiting Gatev and Strahan (2006)

This table reports regressions relating the growth of deposit and nondeposit liabilities to a bank's liquidity demand risk during market stress episodes. Stress is proxied by the commercial paper spread. *Crisis1* is a dummy variable equal to one from 2007Q3 to 2008Q2, and *Crisis2* is a dummy variable equal to one from 2008Q3 to 2009Q2. All specifications are panel regressions with fixed effects for bank organizations and quarterly time dummies. The sample period is from 1990 to 2009, except for the Gatev and Strahan sample (1991 to 2000). Following Gatev and Strahan, the interaction of the three-month Treasury bill rate with the unused commitments ratio is also controlled for. The reported R^2 is within R^2 . The standard errors used in calculating significance levels are clustered at the bank organization level and are shown in parentheses. See the Internet Appendix for variable definitions and details about the bank panel. ***, **, and * indicate 1%, 5%, and 10% significance, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	Δ Deposits/Assets _{<i>t-1</i>}			Δ Nondeposit Liabilities/Assets _{<i>t-1</i>}		
	Gatev and Strahan sample 1991 to 2000	1990 to 2009	1990 to 2009	Gatev and Strahan sample 1991 to 2000	1990 to 2009	1990 to 2009
Unused Commitment Ratio _{<i>t-1</i>}	-0.021*** (0.007)	0.025*** (0.003)	0.025*** (0.003)	0.021*** (0.005)	0.013*** (0.002)	0.016*** (0.002)
Unused Commitment Ratio _{<i>t-1</i>} $\times$ Stress	0.041*** (0.011)	0.007 (0.005)	0.022*** (0.003)	-0.004 (0.003)	0.006* (0.003)	-0.007 (0.003)
Unused Commitment Ratio _{<i>t-1</i>} $\times$ Stress $\times$ Crisis1			-0.039** (0.017)			0.029** (0.011)
Unused Commitment Ratio _{<i>t-1</i>} $\times$ Stress $\times$ Crisis2			0.018 (0.013)			0.014 (0.009)
Unused Commitment Ratio _{<i>t-1</i>} $\times$ Crisis1			0.004 (0.014)			-0.003 (0.009)
Unused Commitment Ratio _{<i>t-1</i>} $\times$ Crisis2			-0.011 (0.007)			-0.018** (0.004)
Controls						
Capital ratio and interactions with stress and crisis	Yes	Yes	Yes	Yes	Yes	Yes
Size and interactions with stress and crisis	Yes	Yes	Yes	Yes	Yes	Yes
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Bank Fixed Effects	114,323	252,526	252,526	114,323	252,526	252,526
Observations	0.05	0.04	0.04	0.02	0.03	0.03
R^2						

(a) Table XI: Deposit and nondeposit liabilities in previous crises

Table XII
The Relationship between Market Stress and Growth of Loans and Total Credit: Was It Different in the 2007 to 2009 Crisis? Revisiting Gatev and Strahan (2006)

This table reports regressions relating the growth of loans and total credit (loans plus commitments) to a bank's liquidity demand risk during market stress episodes. Stress is proxied by the commercial paper spread. *Crisis1* is a dummy variable equal to one from 2007Q3 to 2008Q2, and *Crisis2* is a dummy variable equal to one from 2008Q3 to 2009Q2. All specifications are panel regressions with fixed effects for bank organizations and quarterly time dummies. The sample period is from 1990 to 2009, except for the Gatev and Strahan sample (1991 to 2000). Following Gatev and Strahan, the interaction of the three-month Treasury bill rate with the unused commitments ratio is also controlled for. The reported R^2 is within R^2 . The standard errors used in calculating significance levels are clustered at the bank organization level and are shown in parentheses. See the Internet Appendix for variable definitions and details about the bank panel. ***, **, and * indicate 1%, 5%, and 10% significance, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	Δ Loans/Assets _{<i>t-1</i>}			Δ (Loans + Commitments)/Assets + Commitments _{<i>t-1</i>}		
	Gatev and Strahan sample 1991 to 2000	1990 to 2009	1990 to 2009	Gatev and Strahan sample 1991 to 2000	1990 to 2009	1990 to 2009
Unused Commitment Ratio _{<i>t-1</i>}	0.087*** (0.010)	0.110*** (0.007)	0.113*** (0.007)	-0.009 (0.010)	0.014*** (0.004)	0.023*** (0.004)
Unused Commitment Ratio _{<i>t-1</i>} $\times$ Stress	0.047*** (0.012)	0.004 (0.005)	0.012*** (0.009)	0.031*** (0.011)	-0.035*** (0.006)	0.018* (0.009)
Unused Commitment Ratio _{<i>t-1</i>} $\times$ Stress $\times$ Crisis1			-0.009 (0.018)			0.028 (0.018)
Unused Commitment Ratio _{<i>t-1</i>} $\times$ Stress $\times$ Crisis2			-0.023* (0.013)			-0.029* (0.015)
Unused Commitment Ratio _{<i>t-1</i>} $\times$ Crisis1			-0.004 (0.013)			-0.007*** (0.014)
Unused Commitment Ratio _{<i>t-1</i>} $\times$ Crisis2			-0.021*** (0.005)			-0.033*** (0.005)
Controls						
Capital ratio and interactions with stress and crisis	Yes	Yes	Yes	Yes	Yes	Yes
Size and interactions with stress and crisis	Yes	Yes	Yes	Yes	Yes	Yes
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Bank Fixed Effects	114,323	252,526	252,526	114,323	252,526	252,526
Observations	0.05	0.08	0.08	0.02	0.05	0.05
R^2						

(b) Table XII: Loans and total credit in previous crises

8 Detailed result map

- **Aggregate result.** Deposit inflows into banks were weak before Lehman and surged only after stronger public guarantees.
- **Price result.** Stressed banks and high-commitment banks offered higher deposit rates.
- **Quantity result.** High-commitment banks had weaker deposit growth during Crisis1.
- **Credit result.** High-commitment banks honored drawdowns but cut new credit and developed loan-to-deposit shortfalls.
- **Funding result.** Liquidity-short banks used liquid buffers and government-related borrowing.
- **Solvency result.** Liquidity demand risk is distinct from solvency risk.
- **Historical result.** Previous stress episodes show the traditional bank-liquidity hedge; 2007–2009 reverses it.
- **Policy result.** Liquidity regulation and public liquidity backstops must account for contingent credit commitments.

9 Short summary

- The paper asks whether banks can remain liquidity providers when the banking system itself is in crisis.
- The traditional view says banks provide credit-line liquidity because deposits flow into banks during market stress.
- In 2007–2008, the initial flight to safety went toward explicitly government-backed instruments rather than bank deposits.
- Banks with high unused commitments raised deposit rates but experienced weaker deposit growth.
- These banks honored outstanding commitments but reduced new credit, ran down liquid assets, and borrowed from government-related sources.
- The effects are distinct from solvency risk and are strongest for banks with realized drawdowns.
- Previous crises show the standard liquidity hedge; 2007–2009 shows its breakdown.
- The main lesson is that banks' liquidity-provision role depends critically on confidence in bank liabilities and public backstops.

10 Conclusion

10.1 Core conclusion

Acharya and Mora show that the 2007–2009 financial crisis was not merely a crisis in which banks provided liquidity to borrowers. It was also a crisis of banks as liquidity providers. Banks honored credit lines, but their ability to do so depended heavily on government and government-sponsored liquidity support.

10.2 Economic intuition

Banks normally bundle deposits and commitments because the liquidity demands of depositors and borrowers are not too highly correlated. In a bank-centered crisis, that assumption fails. Borrowers draw credit lines, but depositors do not necessarily provide offsetting funds because they worry about bank risk. The bank's two-sided liquidity business becomes fragile.

10.3 Incremental contribution revisited

The paper's incremental contribution is to show that the liquidity-synergy theory of banking is conditional on the safety of bank liabilities. It takes the Kashyap-Rajan-Stein logic into a systemic crisis environment and demonstrates that the bank deposit-commitment synergy can break down in the aggregate and at the individual-bank level.

10.4 Policy implications

Liquidity regulation should account for contingent commitment drawdowns, not just current balance-sheet liquidity. Deposit insurance and lender-of-last-resort facilities should be priced to reflect systemic liquidity demand risk. Ex post support can preserve credit supply, but it creates fiscal costs and moral hazard, so ex ante liquidity requirements and properly priced public backstops are essential.

10.5 Limitations

The study is observational and U.S.-focused. The bank fixed-effect design is strong, but commitment exposure is not randomly assigned. The exact welfare trade-off between public liquidity support and private credit contraction is not structurally estimated.

10.6 Takeaway

The central takeaway is that banks are not automatically stable liquidity backstops. They are liquidity providers only if depositors trust their liabilities or if public guarantees restore that trust. When banks themselves are at the center of the crisis, deposit-taking and commitment lending can become a source of fragility rather than a hedge.